
BIOLOGY

9700/51

Paper 5 Planning, Analysis and Evaluation

October/November 2019

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **8** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

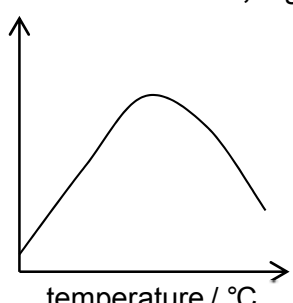
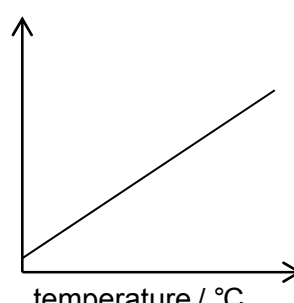
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Question	Answer	Marks
1(a)(i)	as temperature increases, the carbon dioxide increases / increases and then decreases/levels off / AW ;	1
1(a)(ii)	<i>independent variable:</i> temperature ; <i>dependent variable:</i> colour / pH ;	2
1(a)(iii)	any 7 of: 1 <i>ref. to</i> setting up at least 5 water-baths at different temperatures ; 2 <i>ref. to</i> a suitable range of temperatures with units ; 3 same / stated / known, volume / concentration of hydrogencarbonate / indicator (solution) to each test-tube ; 4 <i>ref. to</i> adding oxygen into indicator solution ; 5 <i>ref. to</i> indicator starting at same colour or pH in all test-tubes ; 6 <i>ref to</i> foil, mix, (immediately) start timing / mix, (immediately) foil, (immediately) start timing ; 7 use same / stated / known, volume of <i>Chlorella</i> ; 8 <i>ref. to</i> putting indicator and <i>Chlorella</i> in separate tubes in a water-bath to reach desired temperature / fixed time ; 9 mix <i>Chlorella</i> and indicator and <i>ref. to</i> a specific end point to be measured ; 10 (<i>ref. to</i> a control using), killed / boiled / dead, <i>Chlorella</i> (of same volume) ; 11 (if time / pH) at least three replicates / repeats and finding mean (of time or pH) or identify / eliminate / remove anomalies or (if colour) <i>ref. to</i> using at least three replicates / repeats ; 12 <i>ref. to</i> low risk ;	7

Question	Answer	Marks
1(b)(i)	<u>1</u> time (to reach standard colour / pH) ;	1
1(b)(ii)	temperature on x-axis ; labelled axis and units ; appropriate curve for named axis ; e.g. rate of respiration (s⁻¹ or min⁻¹)  temperature / °C rate of respiration (s⁻¹ or min⁻¹)  temperature / °C	3
1(c)(i)	0.31 ;	1
1(c)(ii)	there is no, (significant) correlation / relationship, between the cell density and the rate of carbon dioxide production or the correlation / relationship between cell density and the rate of carbon dioxide production is not significant ;	1
1(c)(iii)	strong ; positive correlation ;	2

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Question	Answer	Marks
1(c)(iv)	<i>any 2 of:</i> 1 correlation does not mean causation ; 2 fewer than number of paired observations (recommended) for Pearson test ; 3 no replicates / no repeats / only done experiment once ; 4 only narrow range of cell densities ; 5 bacteria / other microorganisms present that are respiring ;	2

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Question	Answer	Marks									
2(a)	any one from: <table><tr><td>tissue</td><td>reason ;</td></tr><tr><td>xylem</td><td rowspan="3">transports water</td></tr><tr><td>epidermis of root</td></tr><tr><td>cortex of root / parenchyma in root / endodermis</td></tr><tr><td>epidermis of leaf</td><td rowspan="2">(too much) water loss</td></tr><tr><td>mesophyll (of leaf)</td></tr></table>	tissue	reason ;	xylem	transports water	epidermis of root	cortex of root / parenchyma in root / endodermis	epidermis of leaf	(too much) water loss	mesophyll (of leaf)	1
tissue	reason ;										
xylem	transports water										
epidermis of root											
cortex of root / parenchyma in root / endodermis											
epidermis of leaf	(too much) water loss										
mesophyll (of leaf)											
2(b)	it is not a 3:1 ratio (expected) ; reason (1 of:) due to interactions between genes ; non resistant plants / certain allele combinations, have reduced chances of survival ;	2									

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Question	Answer	Marks
2(c)(i)	<i>any 3 of:</i> 1 probes / known DNA sequences attached to plate / slide / AW ; 2 <i>ref. to</i> (probe) DNA is single stranded ; 3 <u>mRNA</u> from resistant and non-resistant plants are extracted ; 4 <u>mRNA</u> converted to <u>cDNA</u> ; 5 cDNA / single strand of DNA (from samples) labelled with different colour (fluorescent) dyes (for resistant and non-resistant plants) ; 6 cDNA / single strand of DNA (from sample), allowed to hybridise with probe / added to plates / added to probes ; 7 fluorescence identifies presence of genes ;	3
2(c)(ii)	3 : 1 ;	1
2(c)(iii)	67 / 66.6 ;	1
2(c)(iv)	at 24 hours, genes expressed more rapidly / more genes expressed, in resistant variety compared to non-resistant variety / ora or at 48 hours, less genes expressed, in resistant variety compared to non-resistant variety / ora ; number of genes expressed decreases in resistant variety as time increases ;	2